

APPLICATION OF INTEGRATION

5

1) AREA OF A REGION:

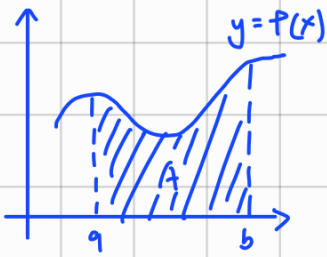
Area of a region under a curve

Area of a the region between two curves

2) VOLUME OF REVOLUTION

Area under the graph

- $\int_a^b f(x) dx$ (above x-axis)



- $\left| \int_a^b g(x) dx \right|$ (below x-axis)

- $\int_a^b f(x) - g(x) dx$

Volume of revolution

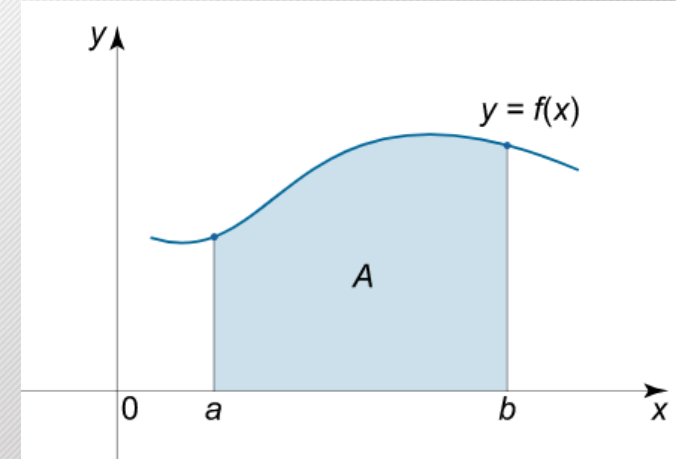
- $V = \pi \int_a^b f(x)^2 dx$

- $V = \pi \int_a^b f(x)^2 - g(x)^2 dx$

Area of a region under a curve (x-axis)

- The area under a curve between two points can be found by doing a definite integral between the two points.
- Figure shows the region bounded by the curve $y = f(x)$, the x - axis and the lines $x = a$ and $x = b$.
- The region located ABOVE x - axis.
- The area of region

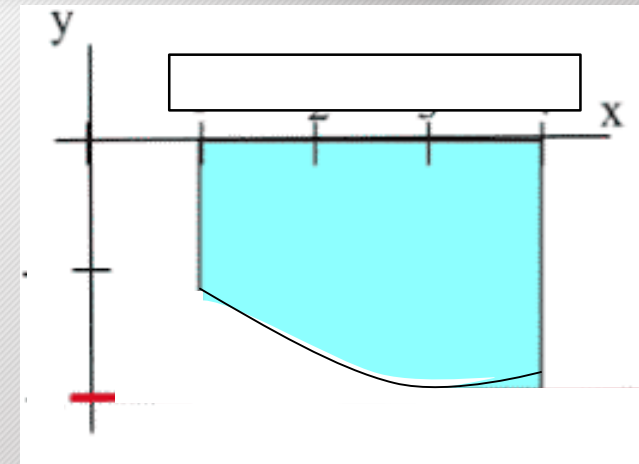
$$\int_a^b f(x) dx$$



Area of a region under a curve (x-axis)

- While for this figure the region located BELOW x -axis. $\rightarrow$ modulus
- The area of region

$$\left| \int_a^b g(x) dx \right|$$



$$1) \quad y = 2 - x^2, \quad x = 0, \quad x = 1$$

$$A_{\text{area}} = \int_0^1 2 - x^2 dx$$

$$= \left[2x - \frac{x^3}{3} \right]_0^1$$

$$= \left(2 - \frac{1}{3} \right) - 0$$

$$= \frac{5}{3} \text{ unit}^2$$

$$2) \quad y = x^2 - 3, \quad x = 1, \quad x = 2$$

$$A_{\text{area}} = \left| \int_1^2 x^2 - 3 dx \right| = \int_{\sqrt{3}}^2 x^2 - 3 dx + \int_1^{\sqrt{3}} x^2 - 3 dx$$

$$= \left| \left[\frac{x^3}{3} - 3x \right]_{\sqrt{3}}^2 \right| + \left[\frac{x^3}{3} - 3x \right]_{\sqrt{3}}^1$$

$$= \left| \left(\frac{8}{3} - 6 \right) - \left(\frac{\sqrt{3}^3}{3} - 3\sqrt{3} \right) \right| + \left(\frac{1}{3} - 3 \right) - \left(\frac{\sqrt{3}^3}{3} - 3\sqrt{3} \right)$$

$$= 0.93$$

$$3) \quad y = \frac{3}{x^2} - 1, \quad x = 1, \quad x = 2$$

$$A_{\text{area}} = \int_1^2 \frac{3}{x^2} - 1 dx$$

$$= \left| \left[-\frac{3}{x} - x \right]_1^2 \right|$$

$$= \left| \left(-\frac{3}{2} - 2 \right) - \left(-3 - 1 \right) \right|$$

$$= 0.58 \text{ unit}^2$$

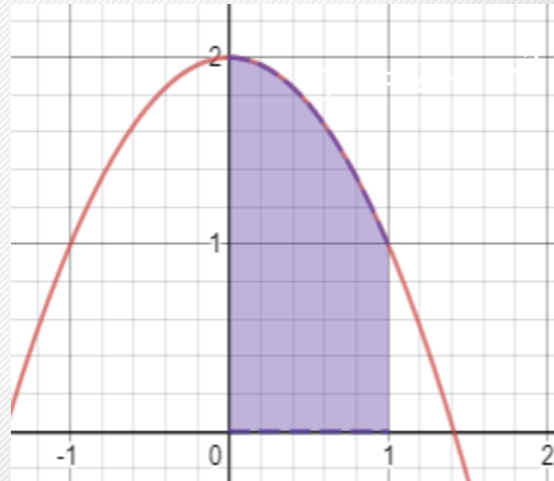
Example

1

- Find the area of the region bounded by the curve $y = 2 - x^2$, the x -axis and the lines $x = 0$, and $x = 1$.

- Solution:

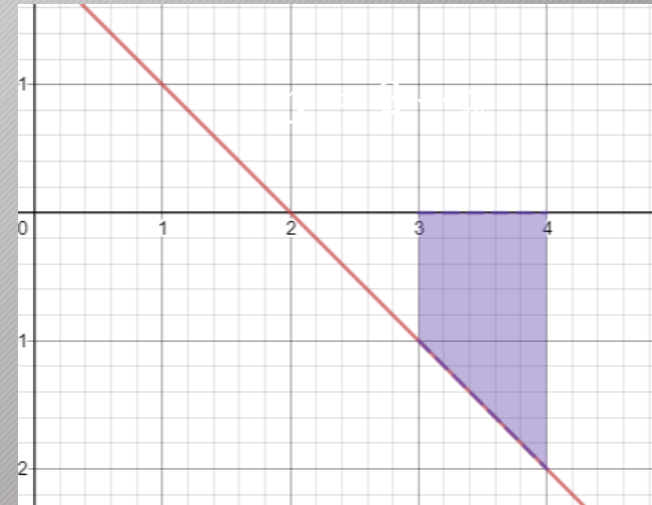
$$\begin{aligned} & \int_0^1 (2 - x^2) dx \\ &= \left[2x - \frac{x^3}{3} \right]_0^1 \\ &= 2 - \frac{1}{3} \\ &= \frac{5}{3} \text{ square units} \end{aligned}$$



- Find the area of the region bounded by the curve $y = 2 - x$, the x -axis and the lines $x = 3$, and $x = 4$.

- Solution:

$$\begin{aligned} & \int_3^4 (2 - x) dx \\ &= \left[2x - \frac{x^2}{2} \right]_3^4 \\ &= \left| 0 - \frac{3}{2} \right| \\ &= \frac{3}{2} \text{ square units} \end{aligned}$$



Example

2

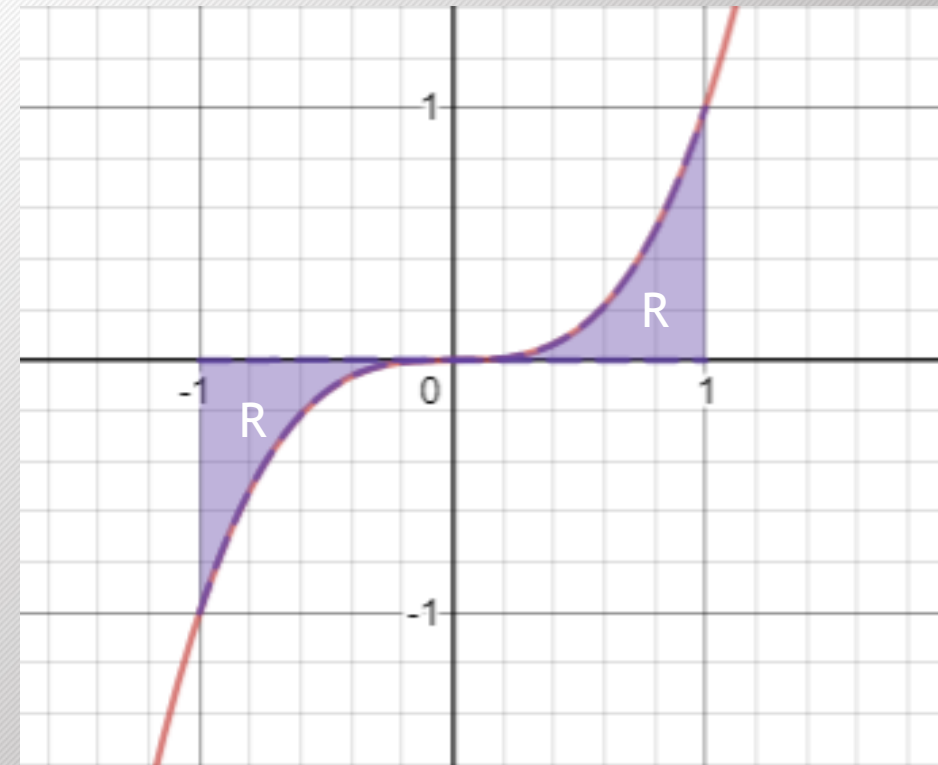
- A region R is bounded by the curve $y = x^3$, the x - *axis* and the lines $x = -1$ and $x = 1$.

- Sketch the region R
- Find the area of region R

Solution:

a)

x	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1
y	-1	$-\frac{1}{8}$	0	$\frac{1}{8}$	1



Continue...

2

- b) Area region above x-axis

$$\begin{aligned} & \int_0^1 x^3 dx \\ &= \left[\frac{x^4}{4} \right]_0^1 \\ &= \frac{1}{4} \end{aligned}$$

- b) Area region below x-axis

$$\begin{aligned} & \left| \int_{-1}^0 x^3 dx \right| \\ &= \left| \left[\frac{x^4}{4} \right]_{-1}^0 \right| \\ &= \frac{1}{4} \end{aligned}$$

Hence , the region of R

$$\int_0^1 x^3 dx + \left| \int_{-1}^0 x^3 dx \right| = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \text{ square units}$$

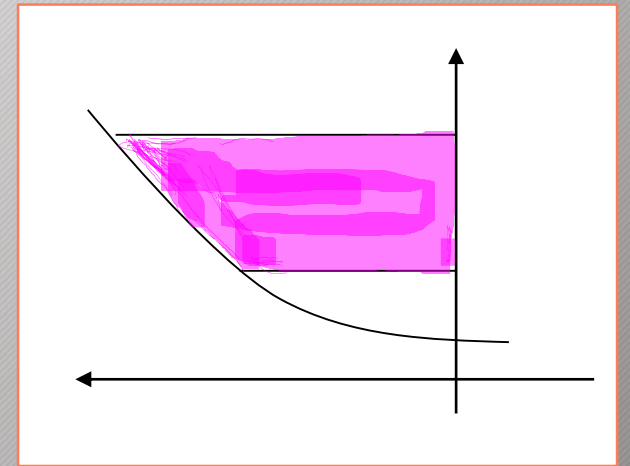
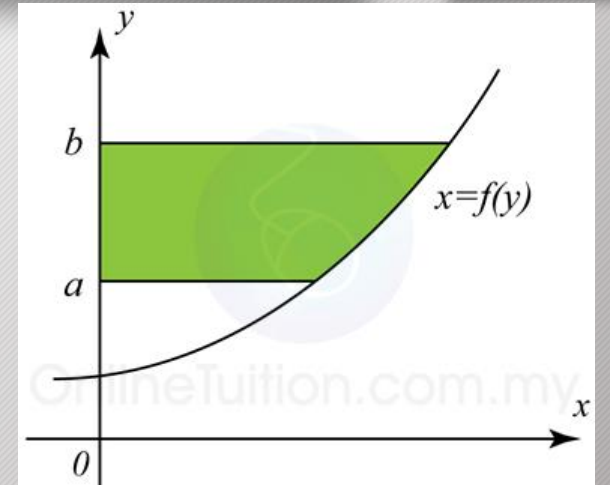
Area of a region under a curve (y-axis)

- Figure shows the region bounded by the curve $x = f(y)$, the y - $axis$ and the lines $y = a$ and $y = b$.
- The region located on the RIGHT y - $axis$.
- The area of region

$$\int_a^b f(y) dy$$

- While for this figure the region located on the LEFT y - $axis$.
- The area of region

$$\left| \int_a^b g(y) dy \right|$$



Example

3

Find the area of the region in the first quadrant bounded by the curve $y = \frac{2}{x}$, the y -axis and the lines $y = 2$ and $y = 4$.

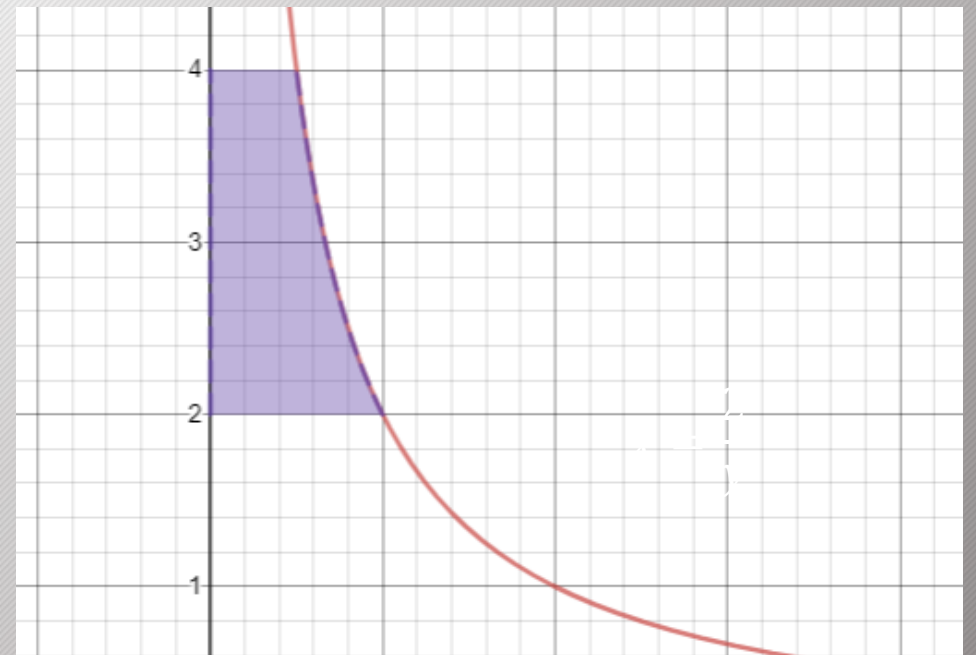
Solution:

1

$$y = \frac{2}{x},$$
$$x = \frac{2}{y}$$

2

$$\int_2^4 \left(\frac{2}{y}\right) dy = [2 \ln y]_2^4$$
$$= 2(\ln 4 - \ln 2) = 2 \ln 2$$
$$= 1.386 \text{ units}^2$$



Example

4

Find the area of the region in the first quadrant bounded by the curve $y = 2 \ln x$, the y -axis, the x -axis and the lines $y = 2$.

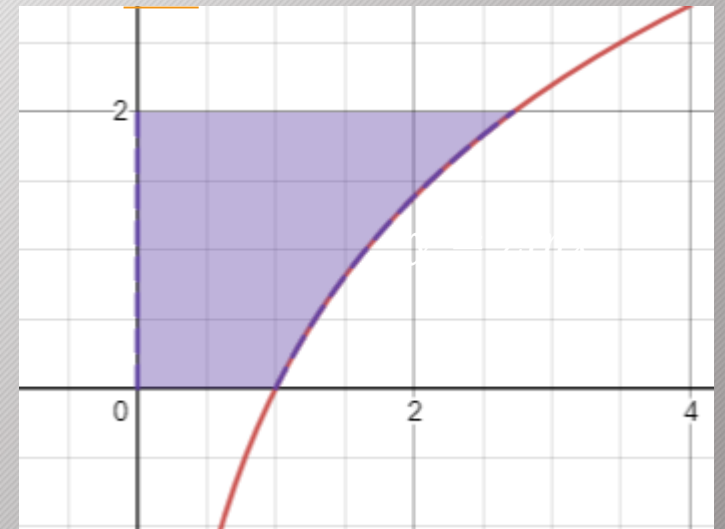
Solution:

1

$$\begin{aligned} y &= 2 \ln x, \\ \ln x &= \frac{y}{2} \\ e^{\ln x} &= e^{\frac{y}{2}} \\ x &= e^{\frac{y}{2}} \end{aligned}$$

2

$$\begin{aligned} \int_0^2 \left(e^{\frac{y}{2}} \right) dy &= \left[2e^{\frac{y}{2}} \right]_0^2 \\ &= 2(e^1 - e^0) \\ &= 3.4366 \text{ units}^2 \end{aligned}$$



Practice 5.1

1. The function f is defined as

$$f(x) = \begin{cases} x + 4k, & x \leq 2 \\ k(x - 3)^2, & x > 2 \end{cases}$$

- (i) Find the value of the constant k so that f is continuous.
- (ii) Sketch the graph of f from $x = 0$ to $x = 4$.
- (iii) Find the area of the region bounded by the curve f , the x -axis and the lines $x = 0$ and $x = 4$.

2. Find the area of the region in the first quadrant bounded by the curve $y = e^{-x}$, the y -axis, the lines $y = 2$ and $y = 4$.

Practice 5.1 (solution)

2. Find the area of the region in the first quadrant bounded by the curve $y = e^{-x}$, the y -axis, the lines $y = 2$ and $y = 4$.

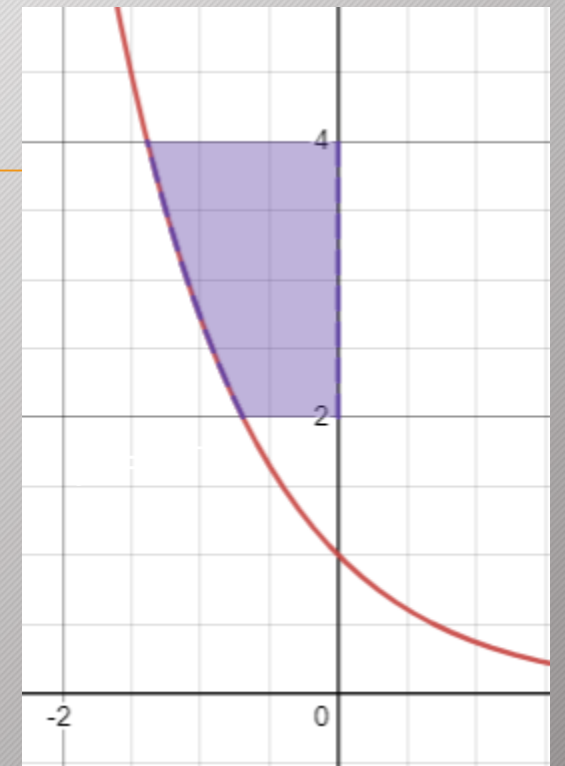
Solution:

1

$$\begin{aligned}y &= e^{-x}, \\ \ln y &= \ln e^{-x} \\ -\ln y &= x\end{aligned}$$

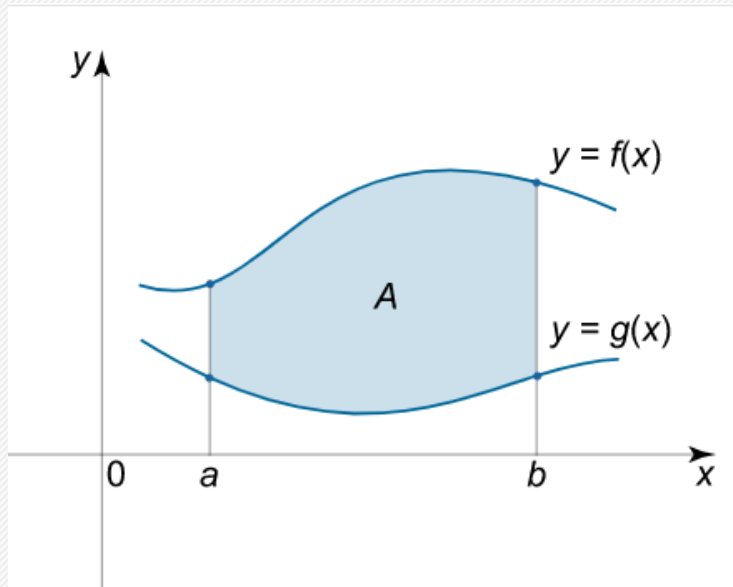
2

$$\begin{aligned}& \left| -\int_2^4 \ln y \, dy \right| \\&= \left| -\left[y \ln y - \int y \cdot \frac{1}{y} dy \right]_2^4 \right| \\&= \left| -[y \ln y - y]_2^4 \right| \\&= \left| -[(4 \ln 4 - 4) - (2 \ln 2 - 2)] \right| \\&= 2.1589 \text{ units}^2\end{aligned}$$

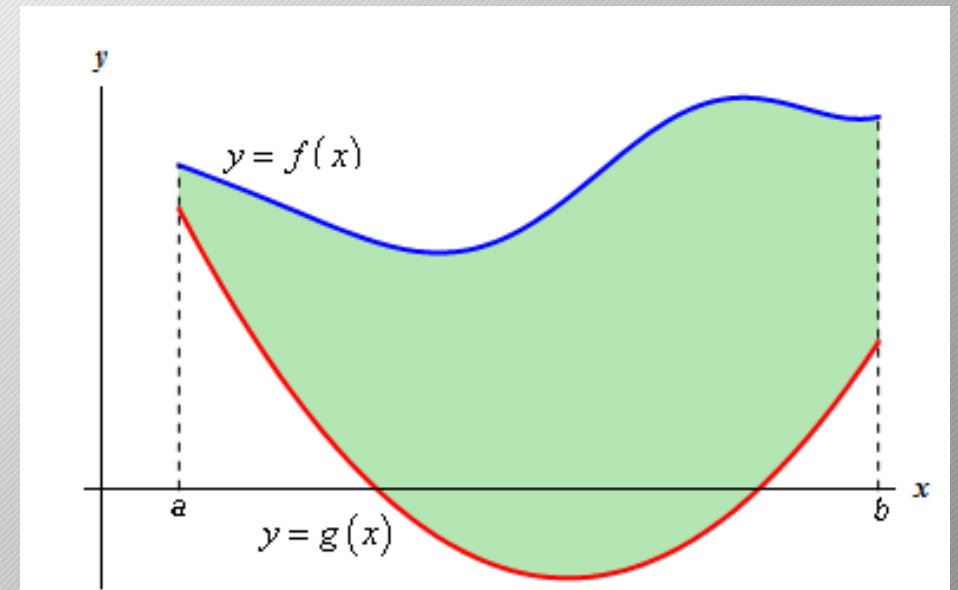


Area of the region Between Two curves

- If $f(x)$ and $g(x)$ are two continuous functions and $f(x) \geq g(x)$ on the closed interval $[a, b]$, then the area between the curves $y = f(x)$ and $y = g(x)$ in this interval is given by

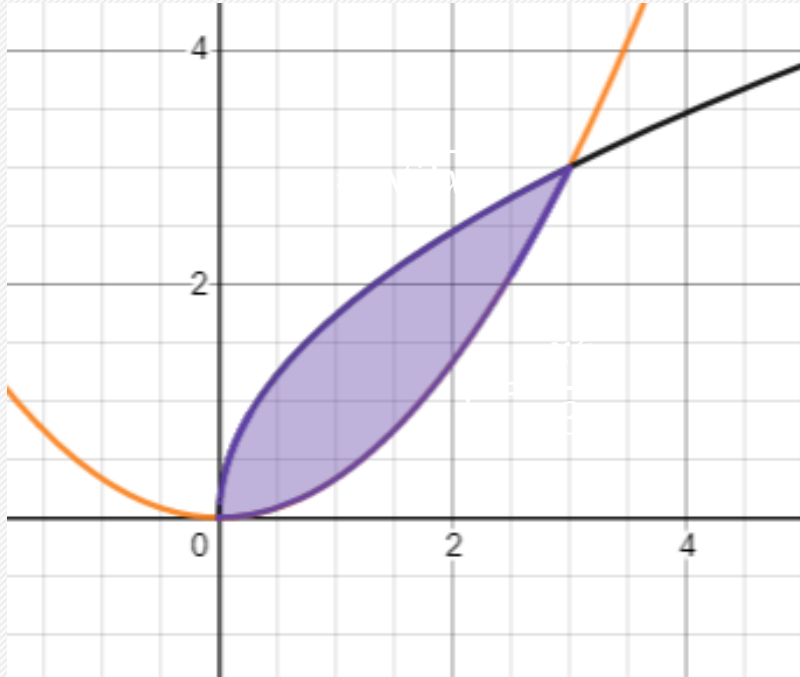


$$\int_a^b [f(x) - g(x)] dx$$



Example

- Find the area of the region bounded by the curve $y = \frac{x^2}{3}$ and the lines $y = \sqrt{3x}$, $x = 0$, $x = 3$.
- Solution:



$$\text{As } \sqrt{3x} > \frac{x^2}{3}$$

$$\begin{aligned}\int_a^b [f(x) - g(x)] dx &= \int_0^3 (\sqrt{3x}) - \left(\frac{x^2}{3}\right) dx \\ &= \left[\sqrt{3} \frac{x^{\frac{3}{2}}}{\frac{3}{2}} - \frac{x^3}{9} \right]_0^3 = \frac{2}{\sqrt{3}} \cdot 3^{\frac{2}{3}} - \frac{27}{9} \\ &= 3 \text{ square units.}\end{aligned}$$

Example

6

- Find the area of the region that bounded by the curve $y = x(x - 1)$ and the line $y = x$.
- Solution:
 1. Upper/ lower limit by finding the intersection point.

$$\begin{aligned}y &= x(x - 1) \\y &= x \\x(x - 1) &= x \\x^2 - x - x &= 0 \\x^2 - 2x &= 0 \\x(x - 2) &= 0 \\x &= 0, x = 2\end{aligned}$$

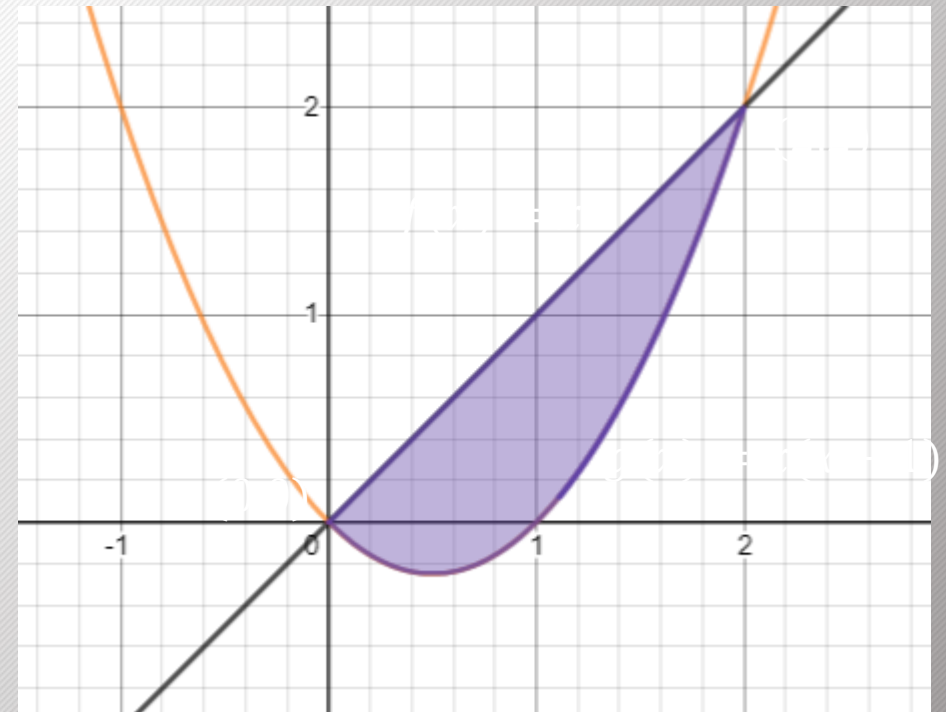
Continue..

6

2. Area of the region

We choose $y = x$ as $f(x)$ and $y = x(x - 1)$ as $g(x)$ so that $g(x) \leq f(x)$ in the interval $[0, 2]$

$$\begin{aligned}\int_a^b [f(x) - g(x)] dx &= \int_0^2 (x - x(x - 1)) dx \\ &= \int_0^2 (2x - x^2) dx \\ &= \left[x^2 - \frac{x^3}{3} \right]_0^2 \\ &= \frac{4}{3} \text{ square units.}\end{aligned}$$

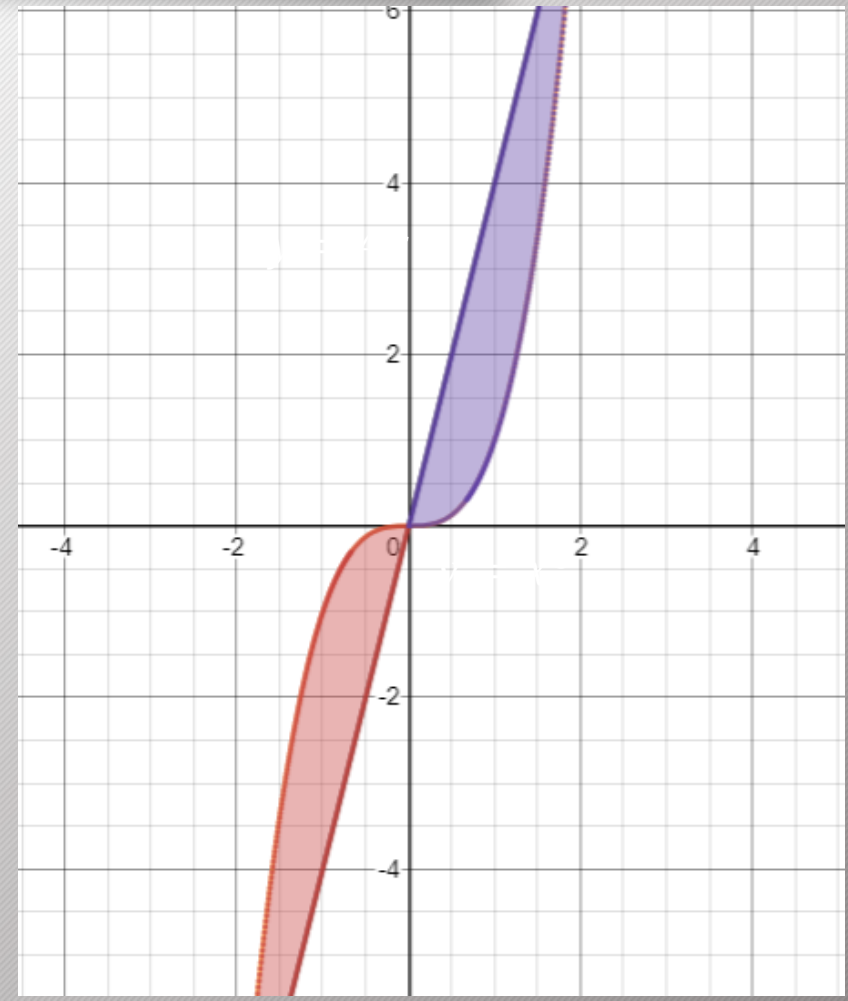


Example

7

- Find the area region bounded by the curves $y = x^3$ and $y = 4x$.
- Solution:
 1. Upper/ lower limit by finding the intersection point.

$$\begin{aligned}x^3 &= 4x \\x^3 - 4x &= 0 \\x(x^2 - 4) &= 0 \\x &= 0, x = \pm 2\end{aligned}$$



Continue..

7

2. Area of the region

(a) Region located **above** x- axis

We choose $y = 4x$ as $f(x)$ and $y = x^3$ as $g(x)$ so that $g(x) \leq f(x)$ in the interval $[0, 2]$

$$\begin{aligned}\int_a^b [f(x) - g(x)] dx &= \int_0^2 (4x - x^3) dx \\ &= \left[2x^2 - \frac{x^4}{4} \right]_0^2 \\ &= 4 \text{ square units.}\end{aligned}$$

Hence , the region of R

$$\int_0^2 (4x - x^3) dx + \int_{-2}^0 (x^3 - 4x) dx = 4 + 4 = 8 \text{ square units}$$

2. Area of the region

(b) Region located **below** x- axis

We choose $y = x^3$ as $f(x)$ and $y = 4x$ as $g(x)$ so that $g(x) \leq f(x)$ in the interval $[-2, 0]$

$$\begin{aligned}\int_a^b [f(x) - g(x)] dx &= \int_{-2}^0 (x^3 - 4x) dx \\ &= \left[\frac{x^4}{4} - 2x^2 \right]_{-2}^0 \\ &= 4 \text{ square units.}\end{aligned}$$

Example

8

- In the figure below find the area of the shaded region bounded by the curve $x = y^2$ and the line $y = x - 2$.

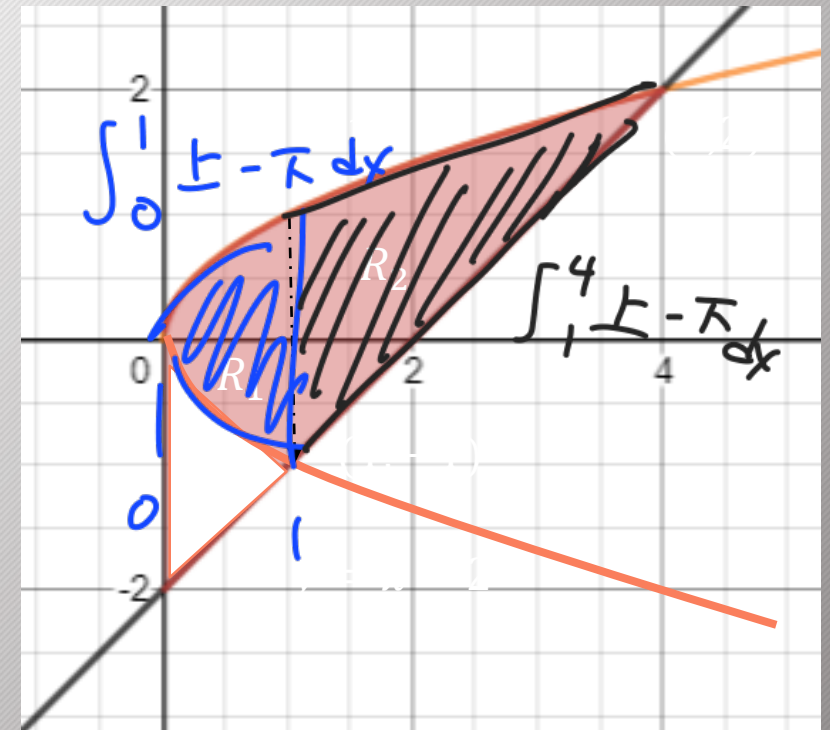
1. By finding the point of intersections:

$$\begin{aligned}x &= (x - 2)^2 \\x^2 - 5x + 4 &= 0 \\(x - 1)(x - 4) &= 0 \\x &= 1, \quad x = 4\end{aligned}$$

The point of intersection is $(1, -1)$ and $(4, 2)$

Hence the integration will be as below

$$\int_1^4 (\sqrt{x} - (x - 2)) dx$$



Continue..

- However the interval is only give the area of region R_2 from $x = 1$ to $x = 4$.
- Therefore we evaluate the area of region from $x = 0$ to $x = 1$ as another region, R_1 .
Upper boundary $y = \sqrt{x}$ and lower boundary $y = -\sqrt{x}$

Area of region R_1

$$\begin{aligned}\int_0^1 (\sqrt{x} - (-\sqrt{x})) dx &= 2 \int_0^1 \sqrt{x} \\ &= \left[\frac{4}{3} x^{\frac{3}{2}} \right]_0^1 = 1 \frac{1}{3}\end{aligned}$$

Area of region R_2

$$\begin{aligned}\int_1^4 (\sqrt{x} - (x - 2)) dx &= \left[\frac{2}{3} x^{\frac{3}{2}} - \frac{1}{2} x^2 + 2x \right]_1^4 \\ &= 3 \frac{1}{6}\end{aligned}$$

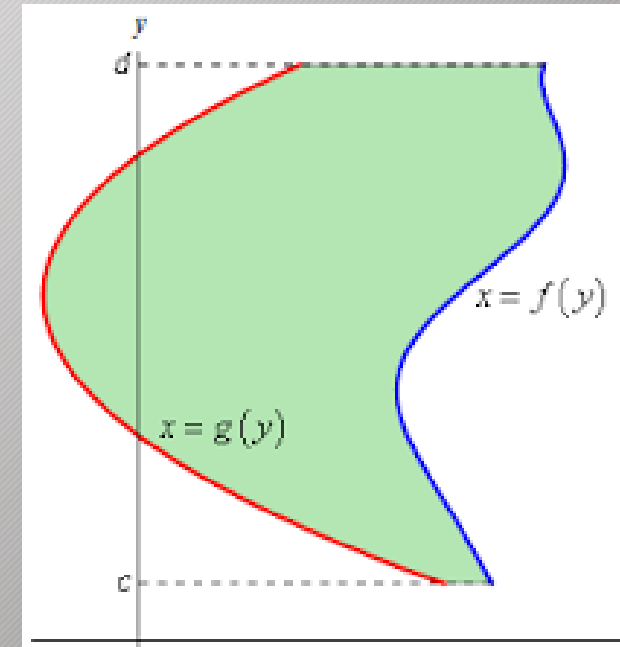
Hence , the region of R

$$\int_0^1 (\sqrt{x} - (-\sqrt{x})) dx + \int_1^4 (\sqrt{x} - (x - 2)) dx = 1 \frac{1}{3} + 3 \frac{1}{6} = 4 \frac{1}{2} \text{ square units}$$

Area between two curves (y -axis)

- In the previous example we obtain the area of the region by integration with respect to x . We need to separate the region and the steps are tedious and complicated.
- We can make it easier by integration with respect to y without dividing the region into few parts.
- The definition is similar with respect to x with the roles of x and y interchanged.
- If $f(y)$ and $g(y)$ are two continuous functions and $f(y) \geq g(y)$ on the closed interval $[c, d]$, then the area between the curves $x = f(y)$ and $x = g(y)$ in this interval is given by

$$\int_d^c [f(y) - g(y)] dy$$



Example

9

- Find the area of the region that bounded by the curve $y^2 = 4x$ and the line $2x + y = 4$.
- Solution:
 - Upper/ lower limit by finding the intersection point.

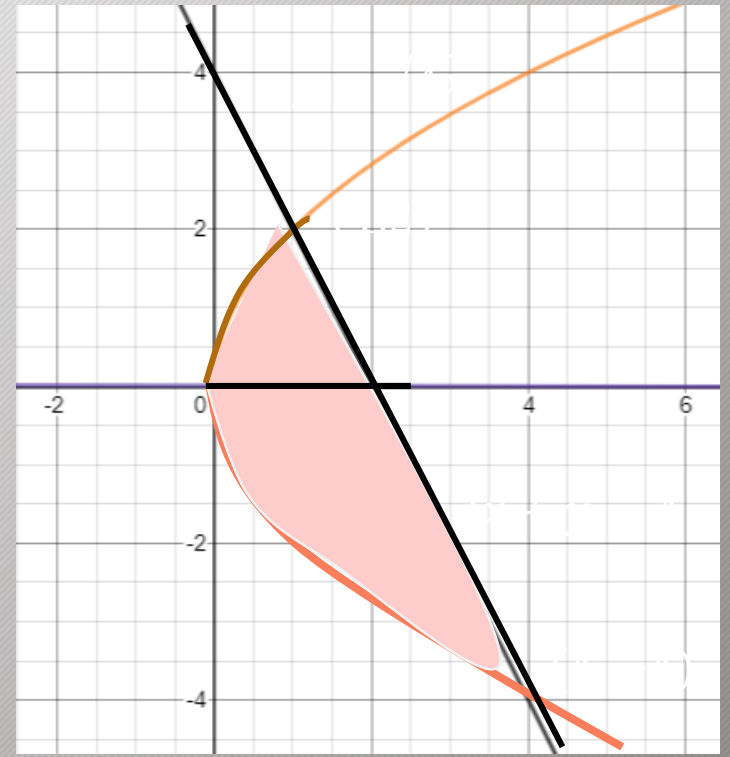
From $2x + y = 4$, we got $y = 4 - 2x$.

Substitute into $y^2 = 4x$, then we will have

$$\begin{aligned}(4 - 2x)^2 &= 4x \\ 16 - 16x + 4x^2 &= 4x \\ 4x^2 - 20x + 16 &= 0 \\ (x - 1)(x - 4) &= 0\end{aligned}$$

when $x = 1, y = 4 - 2(1) = 2$ and

when $x = 4, y = 4 - 2(4) = -4$



Continue..

9

2. Area of the region

We choose $2x + y = 4$ as $f(y)$ and $y^2 = 4x$ as $g(y)$ so that $g(y) \leq f(y)$ in the interval $[-4, 2]$

$$\begin{aligned}\int_a^b [f(y) - g(y)] dy &= \int_{-4}^2 \left(\frac{4-y}{2} - \frac{y^2}{4} \right) dy \\ &= \frac{1}{4} \int_{-4}^2 (8 - 2y - y^2) dy \\ &= \frac{1}{4} \left[8y - y^2 - \frac{y^3}{3} \right]_{-4}^2 \\ &= 9 \text{ square units.}\end{aligned}$$

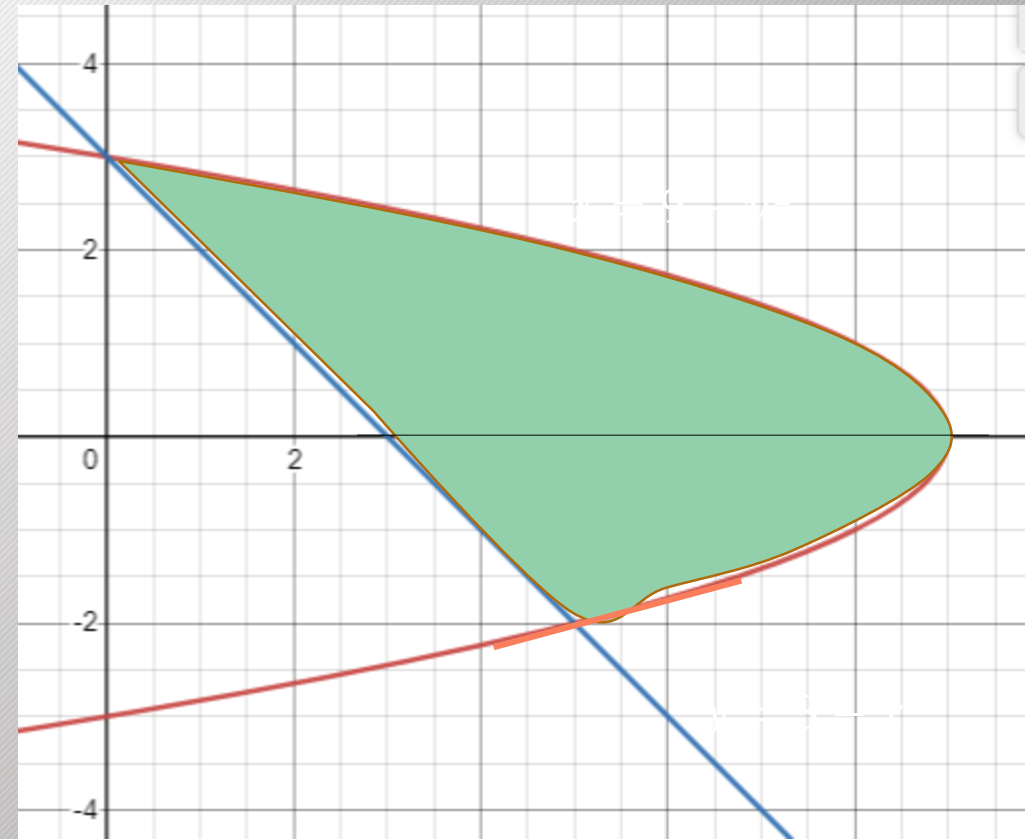
Example

10

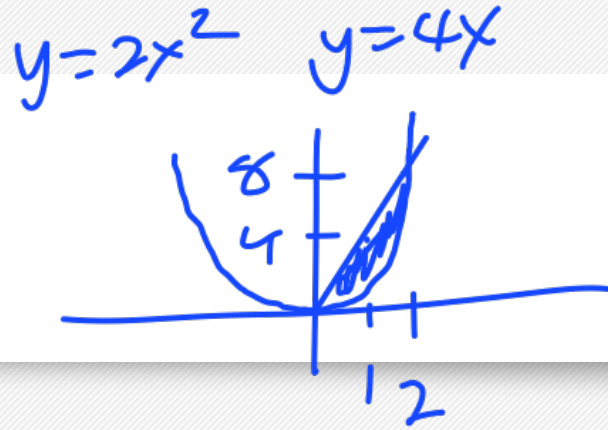
- Region A is bounded by the curve $x = 9 - y^2$ and the line $y = 3 - x$. Find the area of A.

$$\begin{aligned}3 - y &= 9 - y^2 \\ y^2 - y - 6 &= 0 \\ (y - 3)(y + 2) &= 0 \\ y &= 3 \text{ or } y = -2\end{aligned}$$

$$\begin{aligned}\text{Area, } \int_a^b [f(y) - g(y)] dy &= \int_{-2}^3 [9 - y^2 - (3 - y)] dy \\ &= \int_{-2}^3 (6 + y - y^2) dy \\ &= \left[6y + \frac{y^2}{2} - \frac{y^3}{3} \right]_{-2}^3 \\ &= 20.8333 \text{ square units.}\end{aligned}$$



Practice 5.2



$$\begin{aligned}2x^2 &= 4x \\2x^2 - 4x &= 0 \\x^2 - 2x &= 0 \\x &= 0, x = 2\end{aligned}$$

- Given the curve $y = 2x^2$ and the lines $y = 4x$.
 - Find the point of intersection of the curve and the line.
 - Sketch the graph and shade the region enclosed by the curve and the line.
 - Calculate the area of the region described in (b).

$$\begin{aligned}\int_0^2 4x - 2x^2 dx &= \left[2x^2 - \frac{2x^3}{3} \right]_0^2 \\&= \left(8 - \frac{16}{3} \right) - 0 = \frac{8}{3} \text{ unit}^2\end{aligned}$$

Answer:

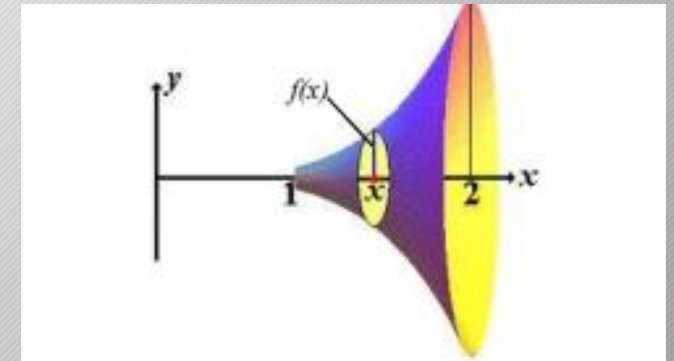
a) (0,0) and (2,8)

c) $\frac{8}{3} \text{ unit}^2$

Volume Of Revolution

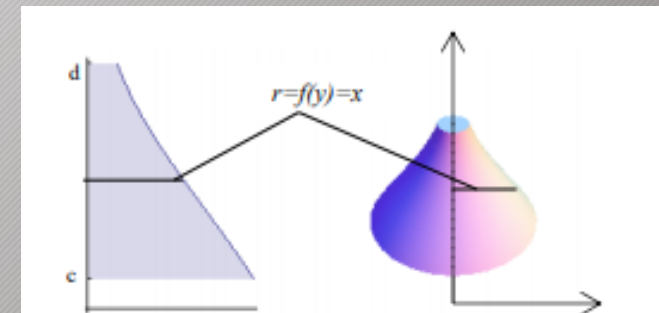
- Now imagine that a curve, for example $y = x^2$, is rotated around the x -axis so that a solid is formed. The volume (V) of a solid generated by revolving the region bounded by $y = f(x)$ and the x -axis on the interval $[a, b]$ about the x -axis through 360° or 2π is

$$V = \pi \int_a^b [f(x)]^2 dx$$



- If the region bounded by $x = f(y)$ and the y -axis on $[c, d]$ is revolved about the y -axis through 360° or 2π is , then its volume (V) is

$$V = \pi \int_c^d [f(y)]^2 dy$$



Example

11

- Find the volume of the solid of revolution when the region bounded by the parabola $y = 2\sqrt{x}$ and x-axis within the interval of $[0,4]$ revolves 360° about x-axis.

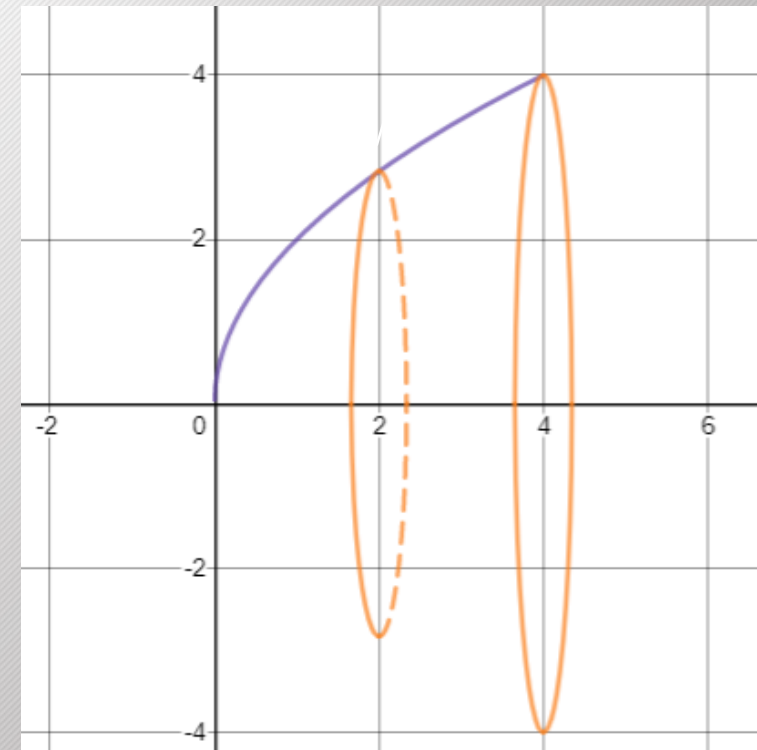
- By using the formula

$$\pi \int_a^b [f(x)]^2 dx = \pi \int_0^4 [2\sqrt{x}]^2 dx$$

•

$$= \pi \int_0^4 4x dx$$

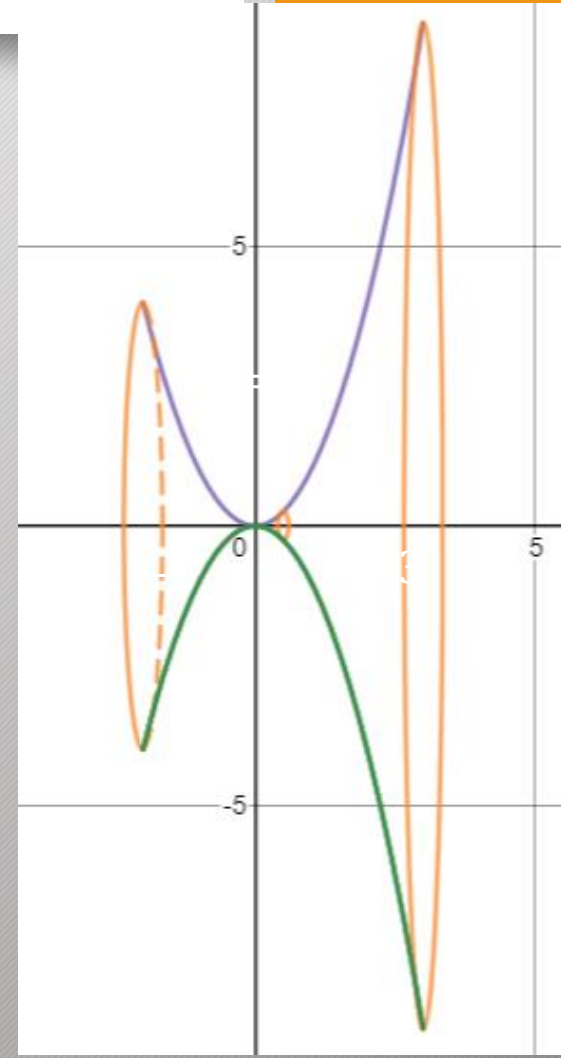
$$= \pi [2x^2]_0^4 = 32\pi \text{ cubic units}$$



Example

- Find the volume of the solid generated by revolving the region bounded by $y = x^2$ and the x-axis on $[-2,3]$ about the x-axis.
- By using the formula

$$\begin{aligned}\pi \int_a^b [f(x)]^2 dx &= \pi \int_{-2}^3 [x^2]^2 dx \\ &= \pi \int_{-2}^3 x^4 dx \\ &= \pi \left[\frac{x^5}{5} \right]_{-2}^3 = 55\pi \text{ cubic units}\end{aligned}$$



Example

13

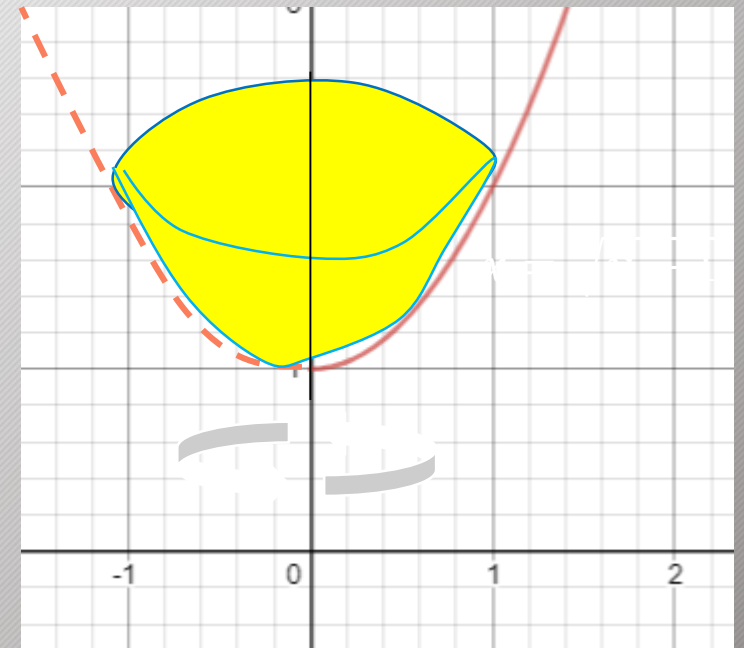
- Find the volume of the solid generated by revolving the region bounded by $y = x^2 + 1$ and the y -axis on $[1,2]$ about the y -axis.
- By using the formula

$$\pi \int_a^c [f(y)]^2 dy = \pi \int_1^2 [\sqrt{y-1}]^2 dy$$

•

$$= \pi \int_1^2 (y-1) dy$$

$$= \pi \left[\frac{1}{2} y^2 - y \right]_1^2 = \frac{\pi}{2} \text{ cubic units}$$



Example

14

- Sketch the region bounded by the curve $y = x^2 + 1$, the x-axis and the ordinates $x = -1$ and $x = 1$. If the region is rotated through 2 radians about x-axis to form a solid of revolution. Calculate the volume of this solid.

Volume Of Revolution

(x-axis / y-axis between two curves)

- If $f(x)$ and $g(x)$ are two continuous and nonnegative functions in the interval $[a,b]$ and $f(x) \geq g(x)$ on the closed interval $[a,b]$, then the volume of revolution when the region bounded by $y = f(x)$ and $y = g(x)$ in $x = a$ and $x = b$, revolves at 360° about the x axis is given by

$$\pi \int_a^b \{[f(x)]^2 - [g(x)]^2\} dx$$

- If $f(y)$ and $g(y)$ are two continuous and nonnegative functions in the interval $[c,d]$ and $f(y) \geq g(y)$ on the closed interval $[c,d]$, then the volume of revolution when the region bounded by $x = f(y)$ and $x = g(y)$ in $y = c$ and $y = d$, revolves at 360° about the y axis is given by

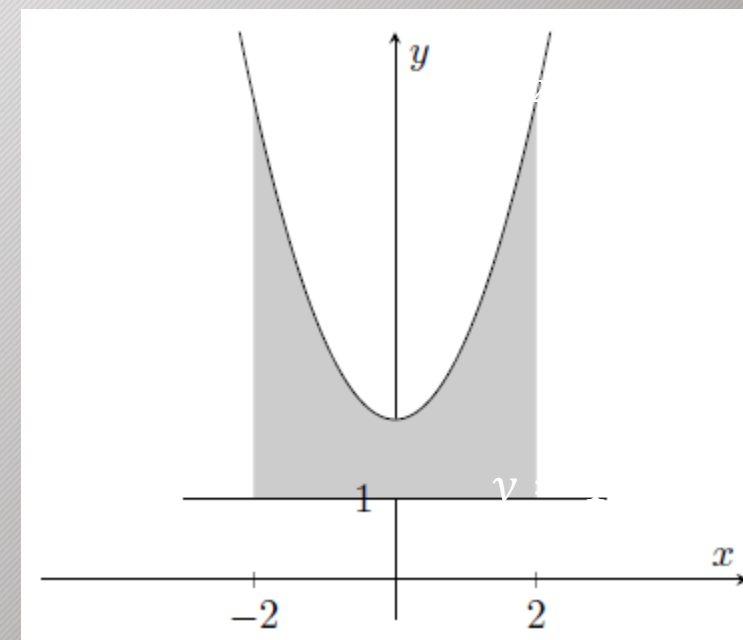
$$\pi \int_c^d \{[f(y)]^2 - [g(y)]^2\} dy$$

Example

15

- Sketch the region that is bounded by the curve $y = x^2 + 2$, lines $x = -2$, $x = 2$, and $y = 1$. Find the volume of solid revolution formed by revolving the region 360 about the x-axis.
- By using the formula

$$\begin{aligned}\pi \int_a^b \{[f(x)]^2 - [g(x)]^2\} dx &= \pi \int_{-2}^2 \{[x^2 + 2]^2 - [1]^2\} dx \\ &= \pi \int_{-2}^2 (x^4 + 4x^2 + 3) dx \\ &= \pi \left[\frac{x^5}{5} + \frac{4}{3}x^3 + 3x \right]_{-2}^2 \\ &= \frac{692}{15} \pi \text{ cubic units}\end{aligned}$$



Example

16

Find the volume of the solid of revolution when the region bounded by $y^2 = 8x$, and $y = x^2$ revolves 360° about y-axis.

1. By finding the intersection point.

$$\sqrt{8x} = x^2$$

$$8x = x^4,$$

$$x(8 - x^3) = 0$$

$$x = 0, x = \pm 2$$

when $x = 2, y = 2^2 = 4$ and

when $x = 0, y = 0^2 = 0$

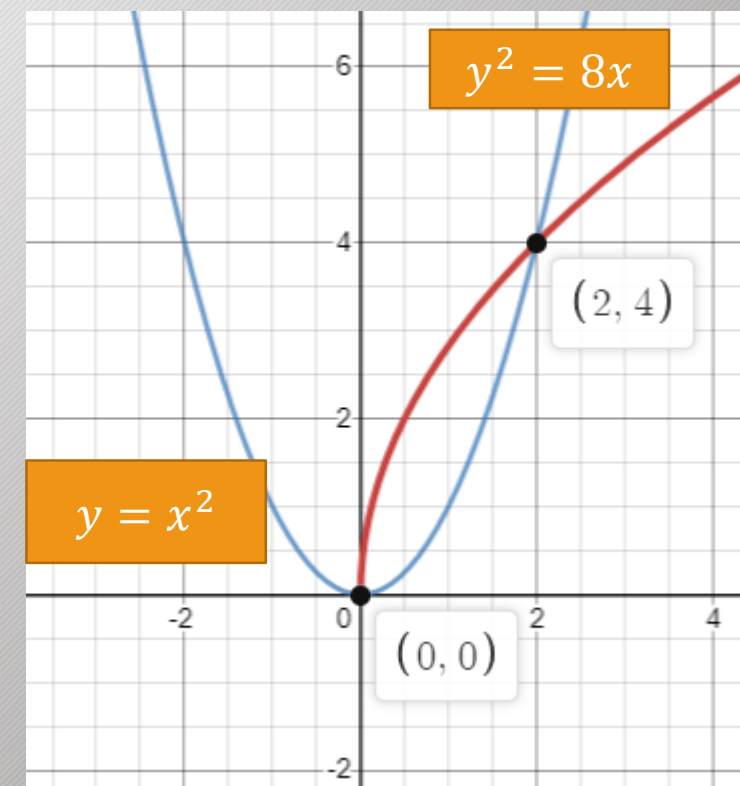
2. By using the formula

$$= \pi \int_0^4 \left\{ [\sqrt{y}]^2 - \left[\frac{y^2}{8} \right]^2 \right\} dy$$

$$= \pi \int_0^4 \left(y - \frac{1}{64} y^4 \right) dy$$

$$= \pi \left[\frac{1}{2} y^2 - \frac{y^5}{320} \right]_0^4$$

$$= \frac{24}{5} \pi \text{ cubic units}$$



Example

17

- Determine the volume of the solid obtained by rotating the portion of the region bounded by $y = \sqrt[3]{x}$, and $y = \frac{x}{4}$ that lies in the first quadrant about the y-axis.

1. By finding the intersection point.

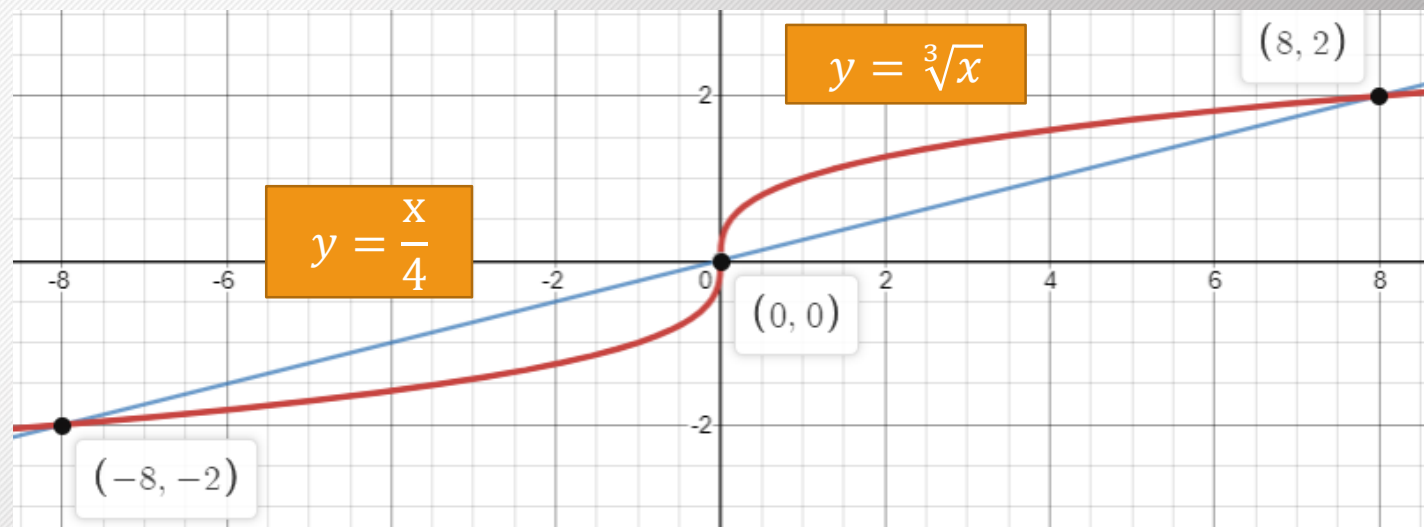
$$\sqrt[3]{x} = \frac{x}{4}$$

$$x = \frac{x^3}{64}, 64x = x^3$$

$$x^3 - 64x = 0$$

$$x(x^2 - 64) = 0$$

$$x = 0 \text{ and } x = \pm 8$$



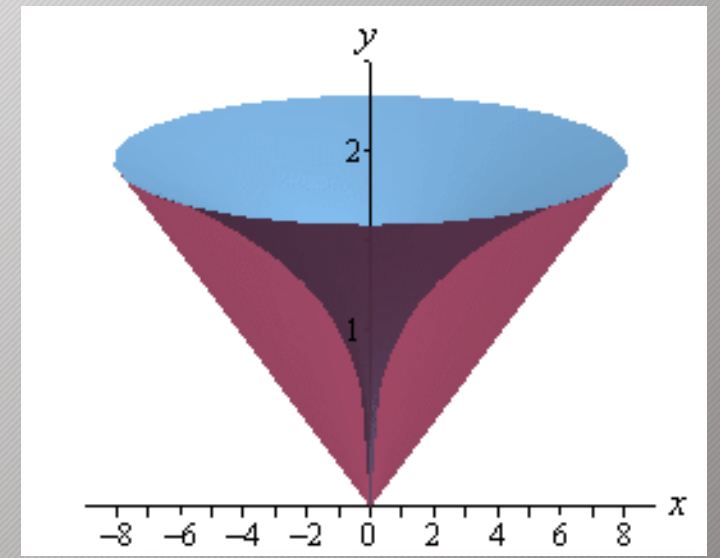
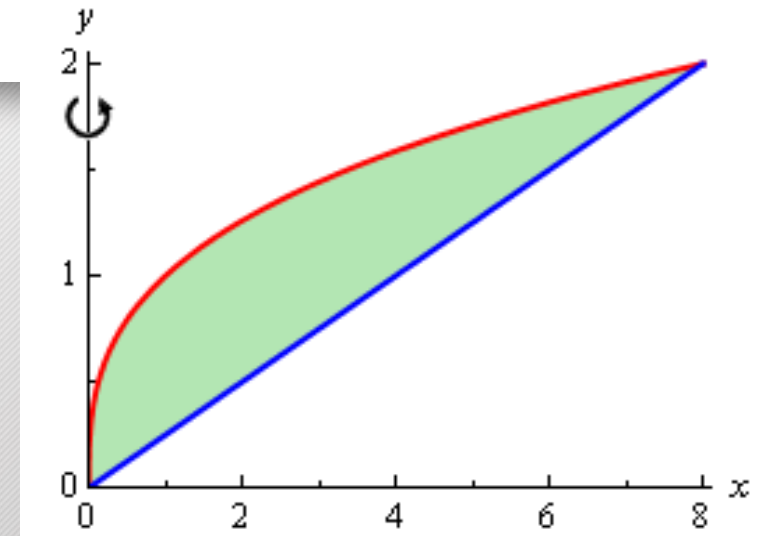
The points of intersection are $(0,0)$, $(8,2)$ and $(-8,-2)$

- There is a portion of the bounding region that is in the third quadrant as well, but we don't want that for this problem.

Continue..

- Therefore the interval is from $y = 0$ to $y = 2$.
2. By using the formula

$$\begin{aligned}\pi \int_c^d \{[f(y)]^2 - [g(y)]^2\} dy &= \pi \int_0^2 \{[4y]^2 - [y^3]^2\} dy \\ &= \pi \int_0^2 (16y^2 - y^6) dy \\ &= \pi \left[\frac{16}{3} y^3 - \frac{y^7}{7} \right]_0^2 \\ &= \frac{512}{21} \pi \text{ cubic units}\end{aligned}$$



Example

18

Find the volume generated when the area in the positive quadrant enclosed between the curve $y = x^2$, lines $y = x + 2$, and revolving the region 360 about the x-axis.

$$\begin{aligned}y &= x^2, y = x + 2 \\x^2 &= x + 2 \\x_1 &= -1, x_2 = 2 \\y &= 1 \quad y = 4 \\(-1, 1), (2, 4)\end{aligned}$$



$$\begin{aligned}V &= \pi \int_{-1}^4 x^2 - x - 2 \, dx \\&= \pi \left[\frac{x^3}{3} - \frac{x^2}{2} - x \right]_{-1}^4 \\&= \left(\frac{28}{3} - \frac{1}{6} \right) \pi \\&= \frac{55}{6} \pi\end{aligned}$$

Practise 5.3

- 1. Sketch the region bounded by the curve $y = x^2 + 2$, lines $x = -2$, $x = 2$ and $y = 1$. Find the volume of solid revolution formed by revolving the region through 2π radians about the x-axis.
- 2. In this figure below, shaded region A is bounded by the curves $y = 4 - x^2$, $y = 3\sqrt{x}$ and x-axis. The curves intersect at a point where a is the x-coordinate.
 - If a is an integer, find value of a .
 - Find the area region of A
 - Find the volume generated when the region A is rotated around x-axis completely.

